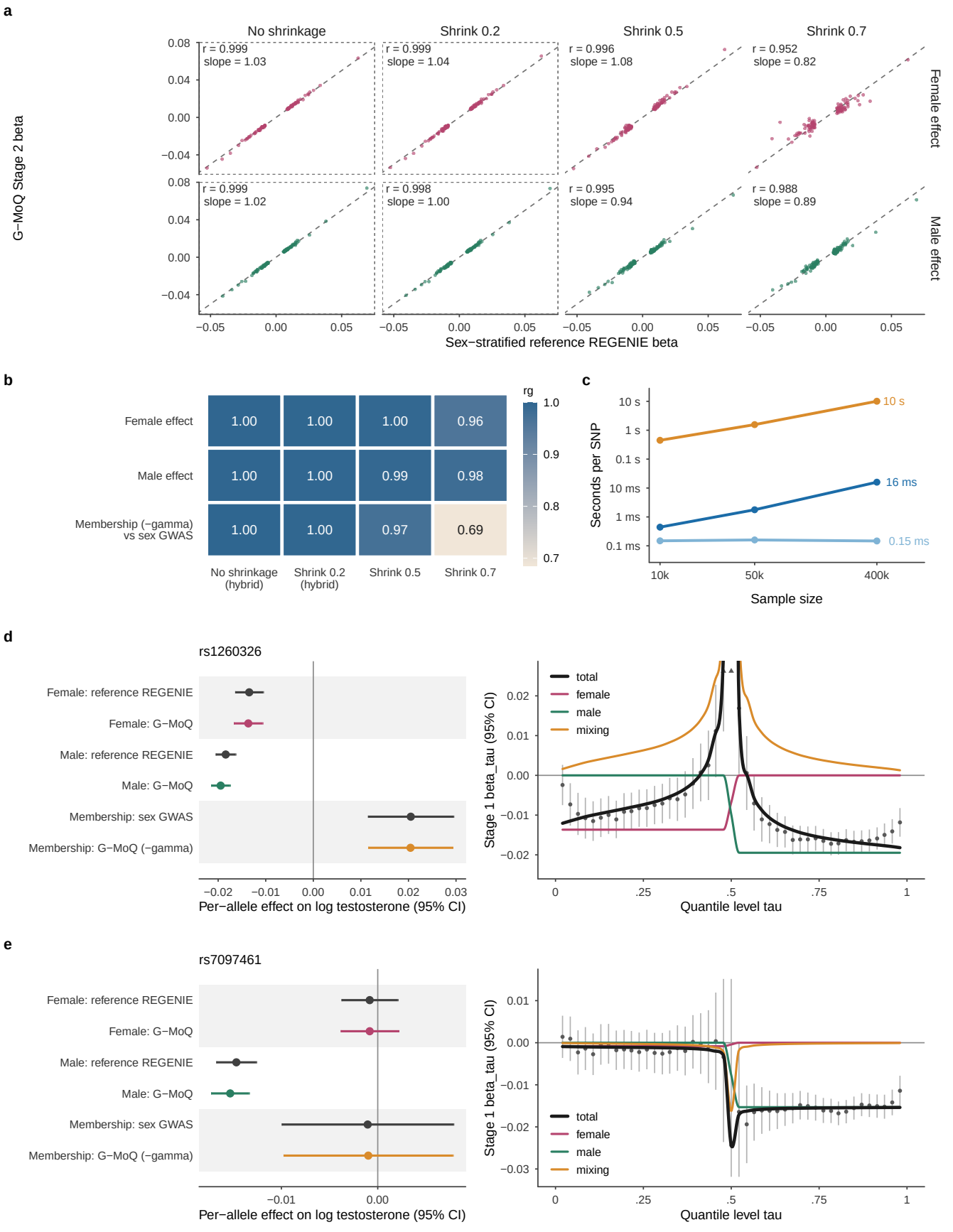

a
No shrinkage
Shrink 0.2
Shrink 0.5
Shrink 0.7
r = 0.999
slope = 1.03
r = 0.999
slope = 1.04
r = 0.996
slope = 1.08
r = 0.952
slope = 0.82
r = 0.999
slope = 1.02
r = 0.998
slope = 1.00
r = 0.995
slope = 0.94
r = 0.988
slope = 0.89
Female effect
Male effect
G−MoQ Stage 2 beta
Sex−stratified reference REGENIE beta
b
Female effect
Male effect
Membership (−gamma) vs sex GWAS
1.00
1.00
1.00
0.96
1.00
1.00
0.99
0.98
1.00
1.00
0.97
0.69
No shrinkage (hybrid)
Shrink 0.2 (hybrid)
Shrink 0.5
Shrink 0.7
rg
c
Seconds per SNP
10 s
1 s
0.1 s
10 ms
1 ms
0.1 ms
10 s
16 ms
0.15 ms
10k
50k
400k
Sample size
d
rs1260326
Female: reference REGENIE
Female: G−MoQ
Male: reference REGENIE
Male: G−MoQ
Membership: sex GWAS
Membership: G−MoQ (−gamma)
Per−allele effect on log testosterone (95% CI)
Stage 1 beta_tau (95% CI)
total
female
male
mixing
Quantile level tau
e
rs7097461
Female: reference REGENIE
Female: G−MoQ
Male: reference REGENIE
Male: G−MoQ
Membership: sex GWAS
Membership: G−MoQ (−gamma)
Per−allele effect on log testosterone (95% CI)
Stage 1 beta_tau (95% CI)
total
female
male
mixing
Quantile level tau